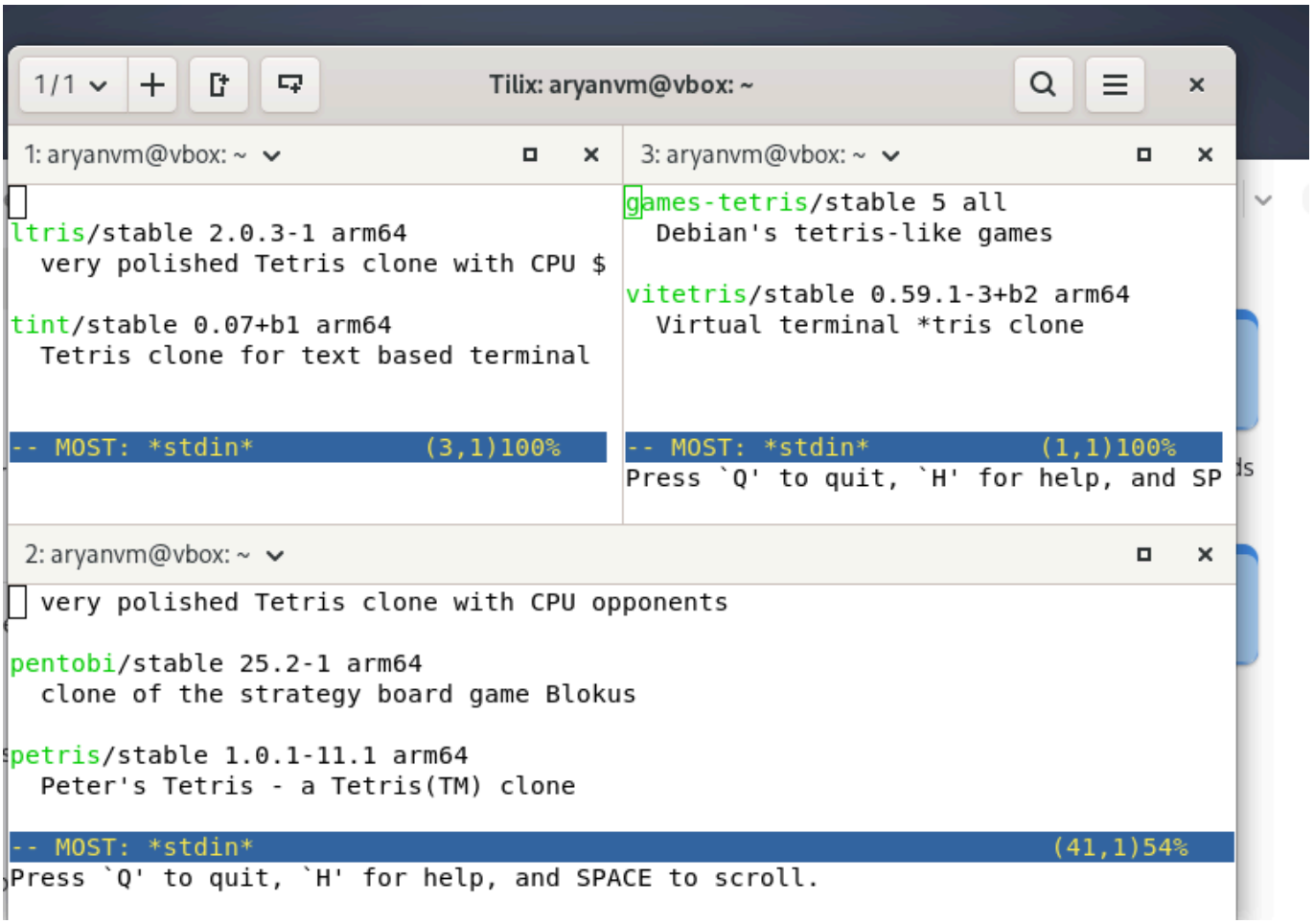
Lab 4 submission

Question 1



1: aryanvm@vbox: ~

```
aryanvm@vbox:~$ sudo apt install blockout2+ cuyo+ blockattack-
[sudo] password for aryanvm:
The following packages were automatically installed and are no longer required:
  libfmt10      libphysfs1          libsdl2-mixer-2.0-0  libxmp4
  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
Use 'sudo apt autoremove' to remove them.

Installing:
  blockout2  cuyo

Installing dependencies:
  cuyo-data  libmikmod3  libsdl-mixer1.2  libsdl1.2debian

REMOVING:
  blockattack

Summary:
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 0
  Download size: 8,041 kB
  Space needed: 4,400 kB / 37.4 GB available

Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main arm64 libmikmod3 arm64 3.3.13-1 [127 kB]
Get:2 http://deb.debian.org/debian trixie/main arm64 libsdl1.2debian arm64 1.2.68-3 [83.1 kB]
Get:3 http://deb.debian.org/debian trixie/main arm64 libsdl-mixer1.2 arm64 1.2.12-19+b1 [81.1 kB]
Get:4 http://deb.debian.org/debian trixie/main arm64 blockout2 arm64 2.5+dfsg1-2 [4,220 kB]
Get:5 http://deb.debian.org/debian trixie/main arm64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main arm64 cuyo arm64 2.1.0-1+b1 [158 kB]
Fetched 8,041 kB in 2s (4,045 kB/s)
(Reading database ... 165828 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:arm64.
(Reading database ... 165801 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_arm64.deb ...
Unpacking libmikmod3:arm64 (3.3.13-1) ...
Selecting previously unselected package libsdl1.2debian:arm64.
Preparing to unpack .../1-libsdl1.2debian_1.2.68-3_arm64.deb ...
```

```
1: aryanvm@vbox: ~
Setting up libmikmod3:arm64 (3.3.13-1) ...
Setting up libsdl1.2debian:arm64 (1.2.68-3) ...
Setting up libsdl-mixer1.2:arm64 (1.2.12-19+b1) ...
Setting up cuyo-data (2.1.0-1) ...
Setting up cuyo (2.1.0-1+b1) ...
Setting up blockout2 (2.5+dfsg1-2) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
aryanvm@vbox:~$ sudp apt purge cuyo
bash: sudp: command not found
aryanvm@vbox:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4
Use 'sudo apt autoremove' to remove them.

REMOVING:
  cuyo*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0
  Freed space: 562 kB

Continue? [Y/n] y
(Reading database ... 166899 files and directories currently installed.)
Removing cuyo (2.1.0-1+b1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
(Reading database ... 166890 files and directories currently installed.)
Purging configuration files for cuyo (2.1.0-1+b1) ...
aryanvm@vbox:~$
```

```
1: aryanvm@vbox: ~
aryanvm@vbox:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4

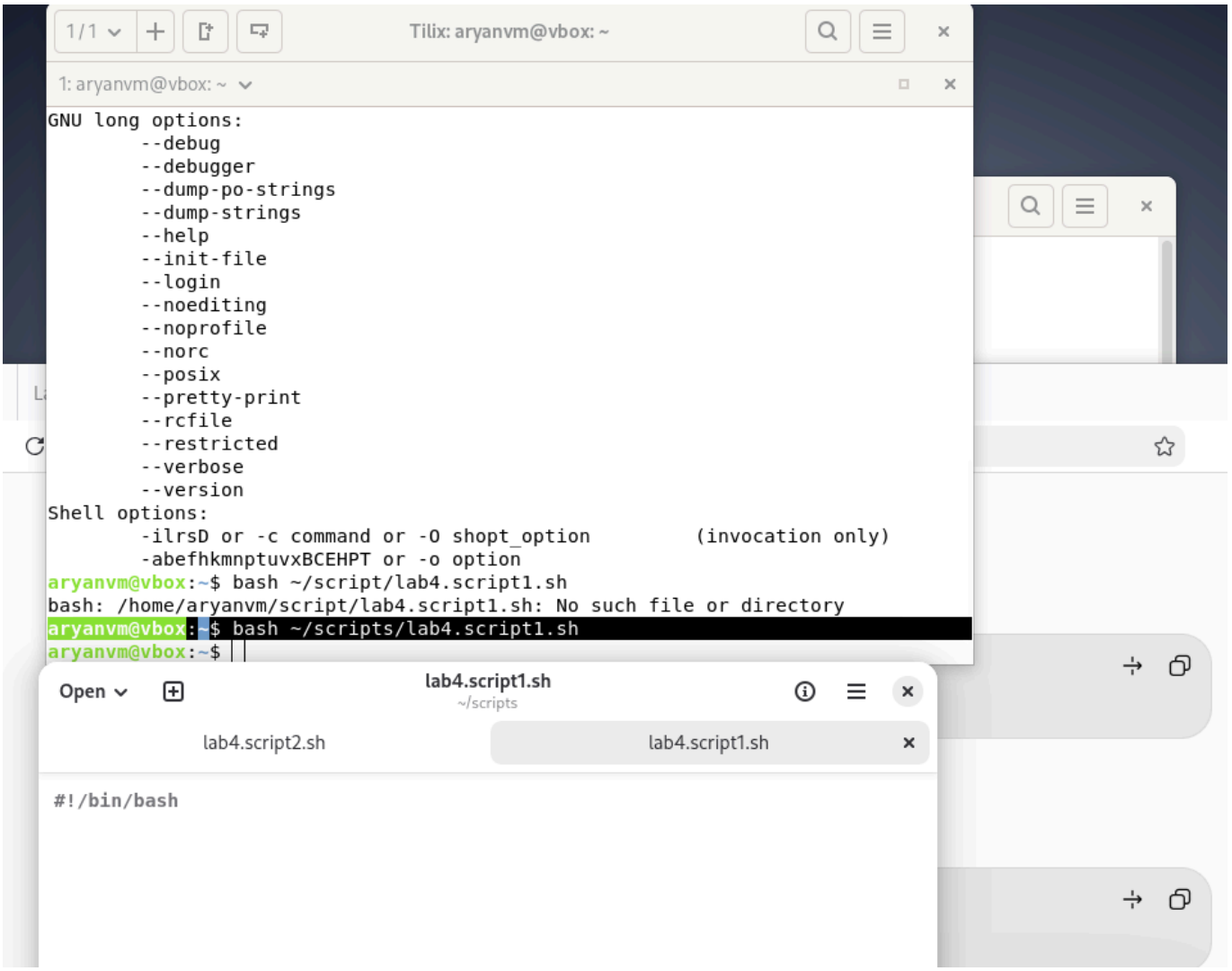
Summary:
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 0
  Freed space: 7,632 kB

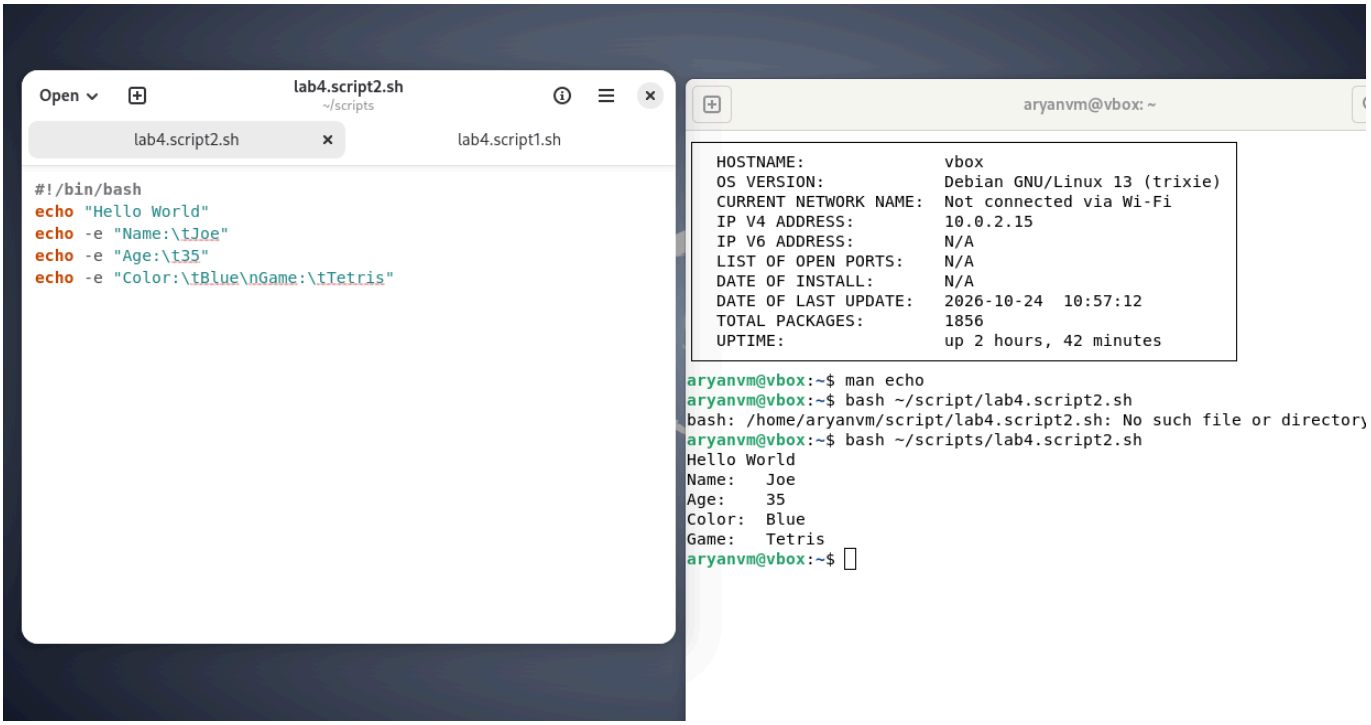
Continue? [Y/n] t
Abort.
aryanvm@vbox:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremove && sudo apt autopurge
REMOVING:
  cuyo-data  libopusfile0  libsdl2-image-2.0-0  libsdl2-ttf-2.0-0
  libfmt10   libphysfs1    libsdl2-mixer-2.0-0  libxmp4

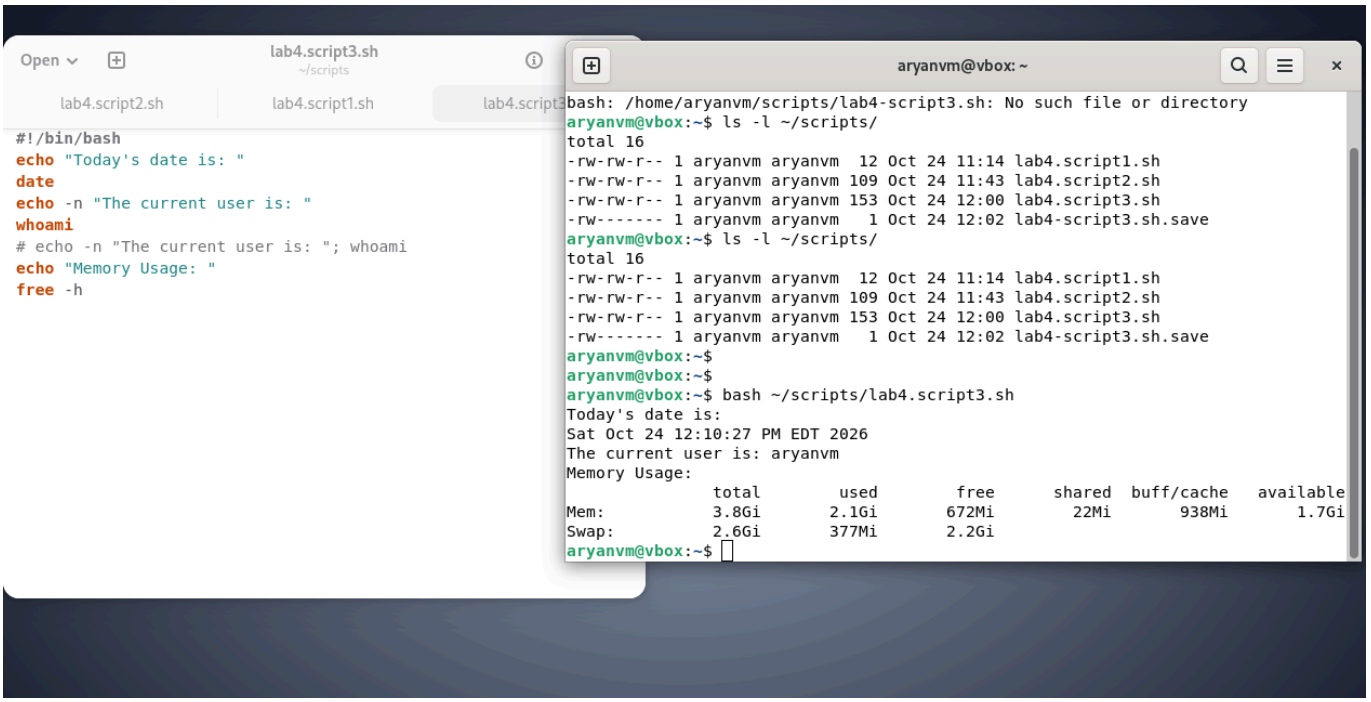
Summary:
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 0
  Freed space: 7,632 kB

Continue? [Y/n] y
(Reading database ... 166890 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:arm64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:arm64 (2.8.1+dfsg-2) ...
Removing libopusfile0:arm64 (0.12-4+b3) ...
Removing libphysfs1:arm64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:arm64 (2.8.8+dfsg-1+deb13u1) ...
Removing libsdl2-ttf-2.0-0:arm64 (2.24.0+dfsg-2) ...
Removing libxmp4:arm64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
aryanvm@vbox:~$
```

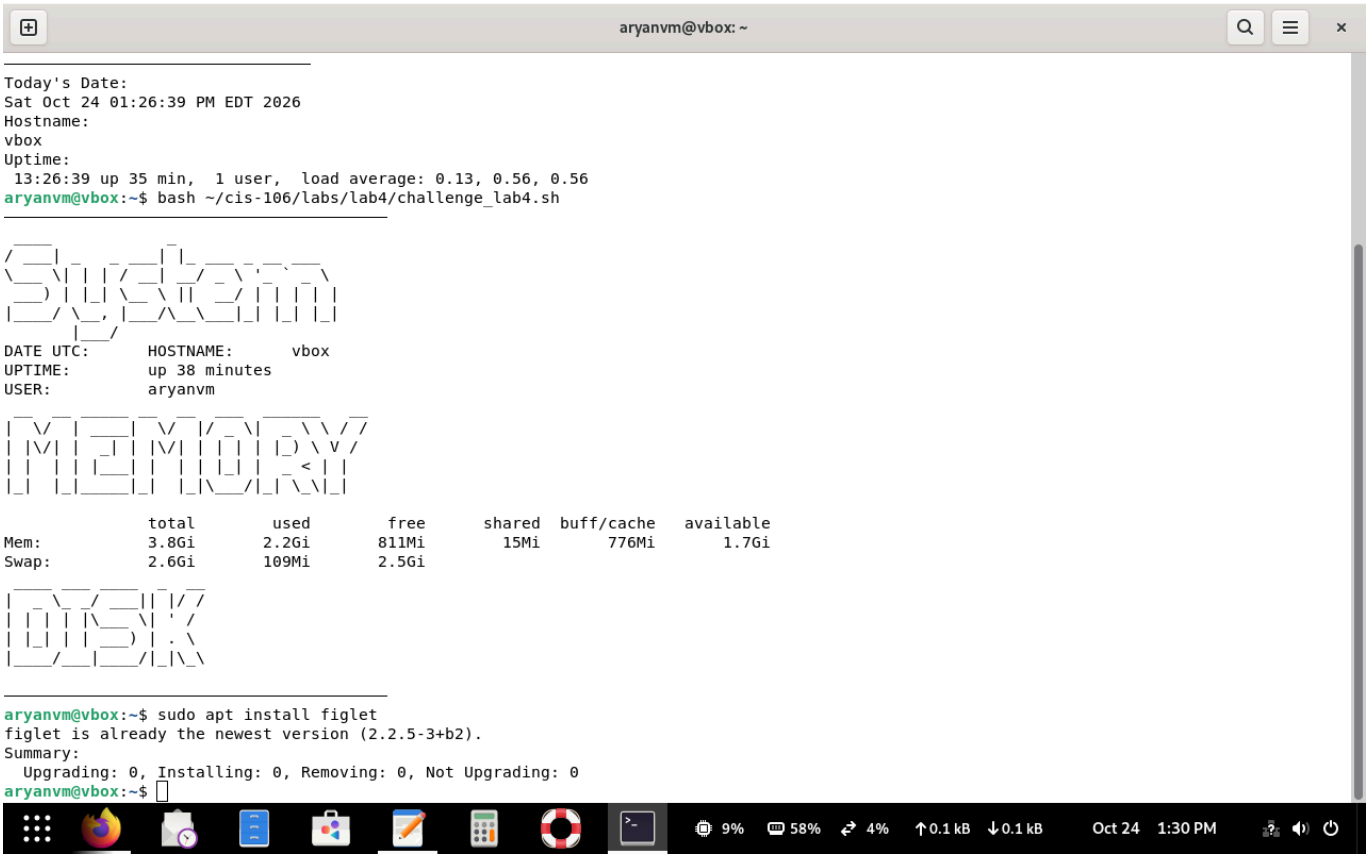
Question 2


Tilix: aryanvm@vbox: ~
1: aryanvm@vbox: ~
GNU long options:
--debug
--debugger
--dump-po-strings
--dump-strings
--help
--init-file
--login
--noediting
--noprofile
--norc
--posix
--pretty-print
--rcfile
--restricted
--verbose
--version
Shell options:
-ilrsD or -c command or -O shopt_option (invocation only)
-abefhkmnptuvxBCEHPT or -o option
aryanvm@vbox:~$ bash ~/script/lab4.script1.sh
bash: /home/aryanvm/script/lab4.script1.sh: No such file or directory
aryanvm@vbox:~$ bash ~/scripts/lab4.script1.sh
aryanvm@vbox:~$
Open
lab4.script1.sh
~/scripts
lab4.script2.sh
lab4.script1.sh
#!/bin/bash


Open
lab4.script2.sh
~/scripts
lab4.script2.sh
lab4.script1.sh
#!/bin/bash
echo "Hello World"
echo -e "Name:\tJoe"
echo -e "Age:\t35"
echo -e "Color:\tBlue\nGame:\tTetris"
aryanvm@vbox: ~
HOSTNAME: vbox
OS VERSION: Debian GNU/Linux 13 (trixie)
CURRENT NETWORK NAME: Not connected via Wi-Fi
IP V4 ADDRESS: 10.0.2.15
IP V6 ADDRESS: N/A
LIST OF OPEN PORTS: N/A
DATE OF INSTALL: N/A
DATE OF LAST UPDATE: 2026-10-24 10:57:12
TOTAL PACKAGES: 1856
UPTIME: up 2 hours, 42 minutes
aryanvm@vbox:~$ man echo
aryanvm@vbox:~$ bash ~/script/lab4.script2.sh
bash: /home/aryanvm/script/lab4.script2.sh: No such file or directory
aryanvm@vbox:~$ bash ~/scripts/lab4.script2.sh
Hello World
Name: Joe
Age: 35
Color: Blue
Game: Tetris
aryanvm@vbox:~$



Question 3



script